Insight from Dataset

Insight is basically what is derived from analyzing dataset. To not break the flow, all of the graphs and plots that may deliver insight are presented here. This also means generally, no EDA (or question 1) will be performed in Tableau. Question 2 will be answered using Tableau, barring few exceptions. Insight or known point will be **bolded**.

Let's start from checking the description of each column, which is not dropped from question 1:

hash_content_id: content_id watched by the play
hash_play_uuid: id of the play
hash_visit_id: session_id of the play
hash_watcher_id: watcher_id of the play
hash_event_id: id of the play (deprecated)
is_login: boolean when user is logged_in when watch
playback_location: where user play the video (direct | embed)
platform: platform used to play
play_time: time the play started
end_time: time the play ended
referrer: attribution of the play, source of play coming from
average_bitrate: average bitrate of play
bitrate_range: bitrate_range set by user to play
total_bytes: bytes used when play
buffer_duration: buffer_duration happen while play content
referrer_group: group of referrer
completed: boolean which indicate the play is complete
player_name: engine player_name
has_ad: indicating the play has ad or not
flash_version:
os_name: os_name of device use to play
os_version: os_version of device use to play
browser_name: browser name of device use to play
browser_version: browser version of device use to play
app_name: app_name used to play
autoplay: indicating the play is autoplay when watched on certain platform
is_premium: indicate the content played is a premium content
app_version: version app being used
play_duration: duration of play
content_type: content type of content (livestreaming | vod | catchup)
stream_type: when content_type is livestreaming, stream_type is used to categorize the livestream (EventStream | TVStream | RadioStream)
title: title of content
category_name: category_name of content

From this list and previous question, we know that several of the column will give different meaning depending on that column have duplicate value or not. For example,

most of the hash column. Usually, sensitive information will be hashed when storing it in the database. These picture from question 1 will show something:

```
df.duplicated(subset=['referrer']).any()
```

```
True
```

```
df.duplicated(subset=['hash_play_id']).any()
```

```
False
```

```
df.duplicated(subset=['hash_content_id']).any()
```

```
True
```

```
df.duplicated(subset=[ 'hash_visit_id']).any()
```

```
True
```

```
df.duplicated(subset= ['hash_watcher_id']).any()
```

```
True
```

```
df.duplicated(subset=['hash_event_id']).any()
```

```
False
```

```
df.duplicated(subset=['hash_film_id']).any()
```

```
True
```

```
df.duplicated(subset=['bitrate_range']).any()
```

```
True
```

```
df.duplicated(subset=['title']).any()
```

```
True
```

We can see that there's no duplicate value in play_id and event_id. From the column description before we know that event_id is deprecated, but this point probably means play_id will replace event_id. On that note, other hash and another columns have duplicate value.

This means there are people that come back for viewing or repeat views. There's also show or program that repeatly accessed compared to the other items.

Look at the play_id. Despite the point above, play_id is unique. Which means each separate play will be recorded differently, regardless of repeat viewer or viewing. This is actually a common practice when storing item receipts, so it can be easily identified. Continuing from question 1 results, we still have the correlation value. Based from my experience, high correlation value start from 0.5 or -0.5. The closer they are to 1 or -1, means better correlation value. Remember, the correlation value in question 1 mostly didn't fulfill this condition.

Except probably one instance. Let's just check the correlation table again.

```
df.corr()
```

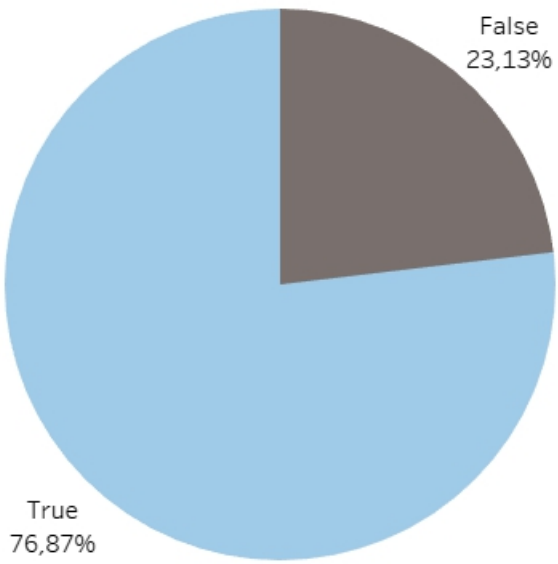
	is_login	average_bitrate	total_bytes	buffer_duration	has_ad	is_premium	play_duration
is_login	1.000000	0.319692	0.101245	-0.001672	-0.231610	0.167211	0.158977
average_bitrate	0.319692	1.000000	0.120405	0.002231	-0.105269	0.053275	0.069359
total_bytes	0.101245	0.120405	1.000000	0.002588	-0.154921	0.012104	0.812459
buffer_duration	-0.001672	0.002231	0.002588	1.000000	-0.005586	-0.000286	0.001663
has_ad	-0.231610	-0.105269	-0.154921	-0.005586	1.000000	-0.168976	-0.129162
is_premium	0.167211	0.053275	0.012104	-0.000286	-0.168976	1.000000	0.062105
play_duration	0.158977	0.069359	0.812459	0.001663	-0.129162	0.062105	1.000000

We can see in the table that is_login and average_bitrate have quite score, 0.319692. However, there's even higher score, which is between play_duration and total_bytes. Their correlation value is a whooping 0.812459. We can concur that play_duration and total_bytes are positively related.

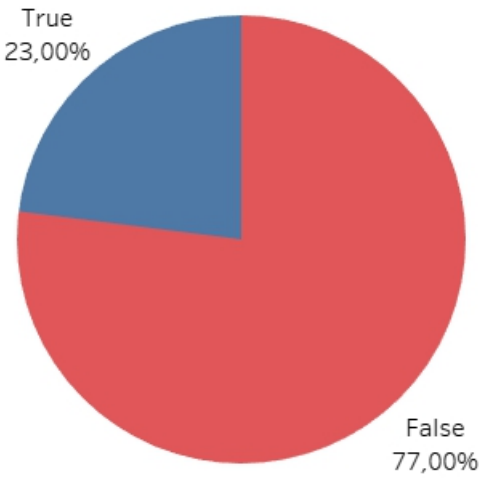
This means that if either play_duration or total_bytes are up, then the other must be rising too. And vice versa.

On another note, we can see something amiss in the correlation table. Column has_ad and is_premium supposedly only have 2 possible input, either TRUE or FALSE. Why is it possible they are included in the table? Yes, you're right. They automatically read as a boolean expression, and auto-translated it to 1 and 0. Thus, they eligible for correlation table. This actually one of the advantage using either Tableau or Jupyter Notebook for data analytics, as they can automatically read dates and boolean expression.

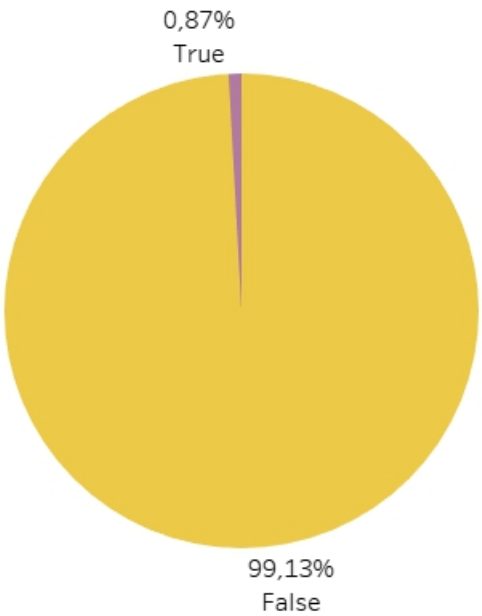
Moving on, let's continue digging for insight. As has_ad and is_premium is already discussed earlier, why don't we do them instead? Using Tableau, I created a pie chart with has_ad values.



This seems pretty uneven. It's clear from the chart that most of the video that watched have ads. True value have 82.107 counts while False have 24.701 counts. So what can be inferred from this? Let's not jump into conclusion, and check the other binary column first. Next, we check is_login and is_premium. Both are using pie chart.

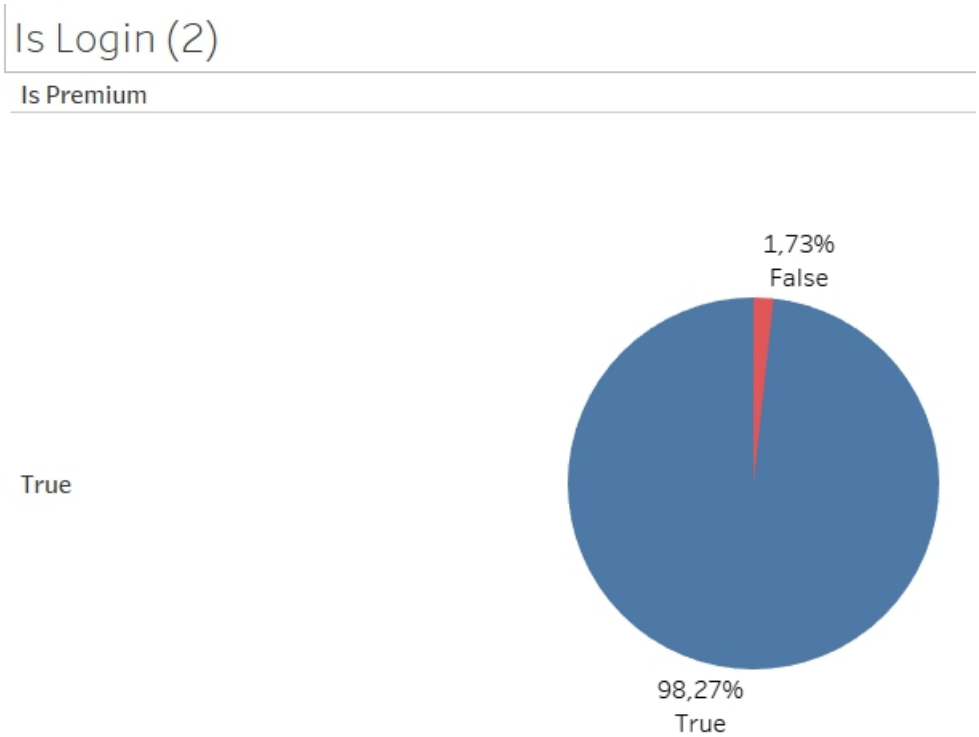


Above chart is the column is_login. We can also see the chart is very uneven. False value is 82.244 while the True value is 24.564. The majority of user is not logged in to, I assume, to Vidio app. Then a question arise. So the majority of user is not watching directly from the app? If so, from where they are watching? Fret not, the question will be answered later. This also ties nicely with question 4, possibly even influencing it somehow. Before we can answer those question, let's see the chart for is_premium.



Looking at this chart, almost tempted me to call it Pacman. This chart isn't even uneven (heheh). It's like one pie is eating another. The True value percentage is like a

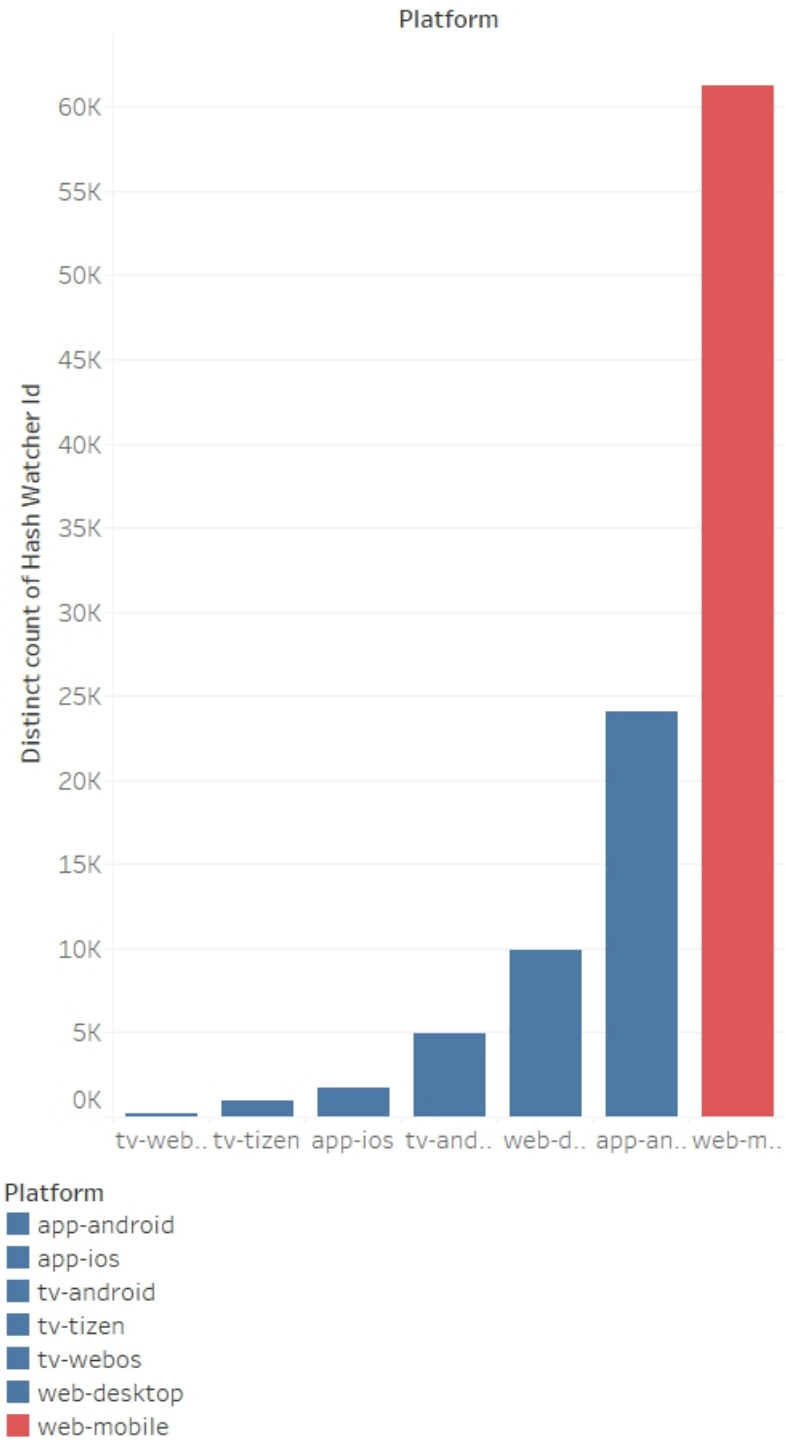
needle in the cheese. They only have 925 counts, while the False value have 105.883 counts. What we can infer from this is almost all of the viewer don't watch premium content, which I assume is must-pay-per-view. And due to it being premium content, I concluded that the viewer watching it probably has True in their is_login. Let's check it with chart.



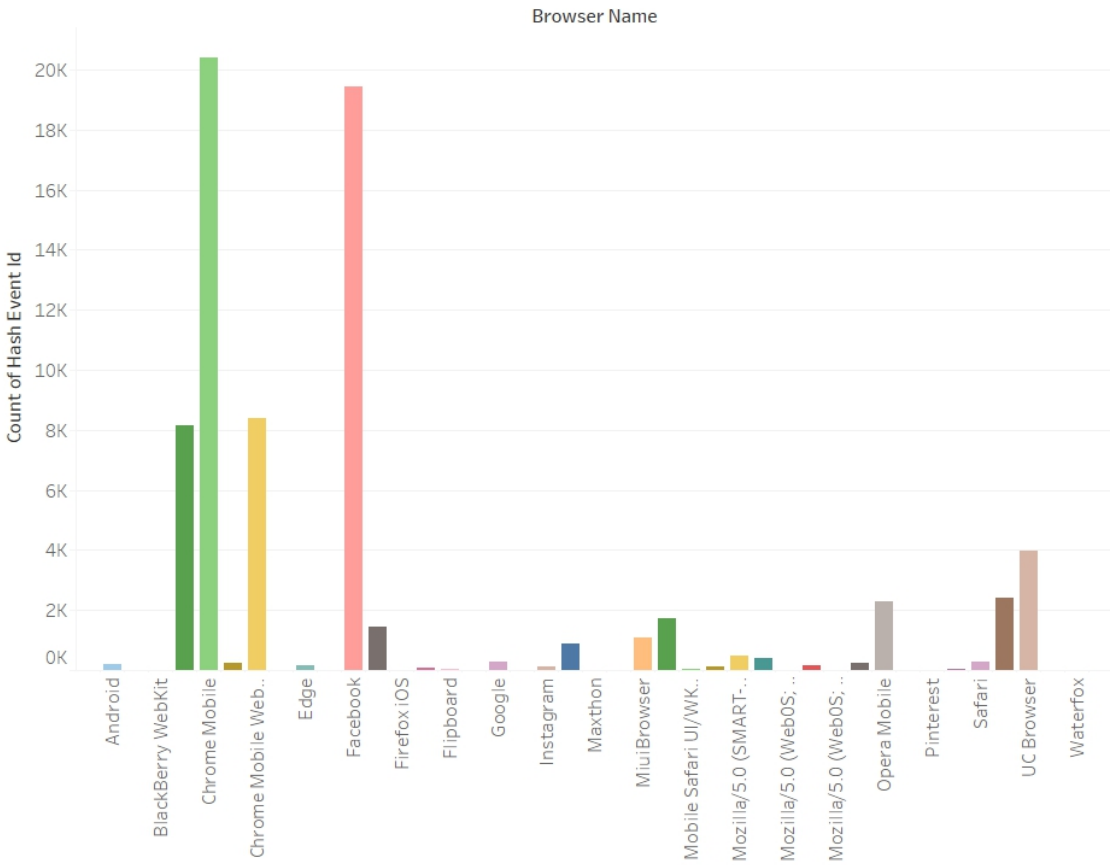
There's still some viewers that didn't login despite watching premium content, which kinda not making any sense. Unless they pirate or using haxx however, without login they should not able to access premium content. Regardless, most of them viewers are logged.

We can conclude that the majority of viewers are watching for more free content, even if they are registered and logged on, and that is why there are many ads in it.

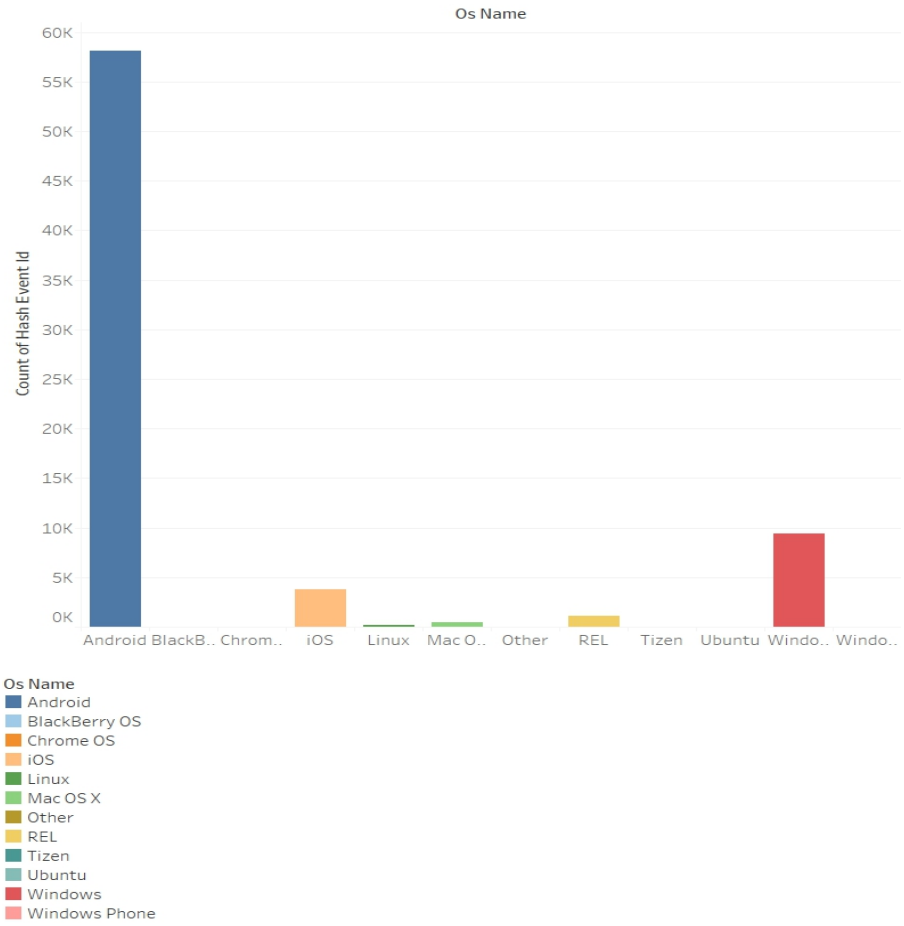
Now we will investigate it further. The previous unanswered question, from where the viewers are watching, isn't yet answered. We will see the chart of platform first. Coincidentally, this chart will be related to question 4. It will have more detailed explanation in the answer, so let's just see how many platform people watch in the dataset and what platform have the highest numbers.



We can see that the highest is web-mobile. It also have a significant gap from the second place. **This means the majority of the viewer watch the videos from web-mobile platform**. This chart probably has a relation to browser name. Logically, if the number one platform is web-mobile then the reigning top browser is related to mobile. We have two chart that is need to be checked, the browser name and the OS name. In case you don't get it yet, OS probably will have a top number related to mobile device, so we have to check it for more info.

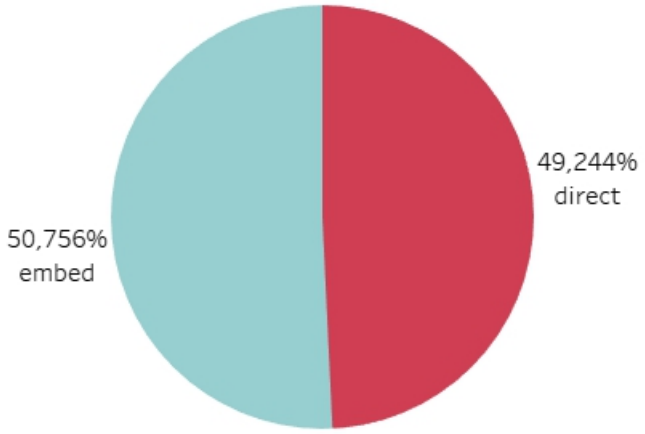


Below is the graph for OS Name.



After comparing those 2 charts, we can draw several conclusions. **Android domination towards the other OS means the overwhelming majority of viewers used Android. This also means most of them accessed the video from their mobile phone.**

This is inline with the chart about platform, where the first and second position is from web-mobile and app-android respectively. It isn't surprising that most of the numbers come from mobile devices. This also explain something quite unusual in the browser name chart. Since when Facebook is a browser? The answer is because people actually watch embedded video while opening Facebook, maybe using app or the web. To confirm this, let's check the chart for playback position, is there a significant amount of either playback or embed?

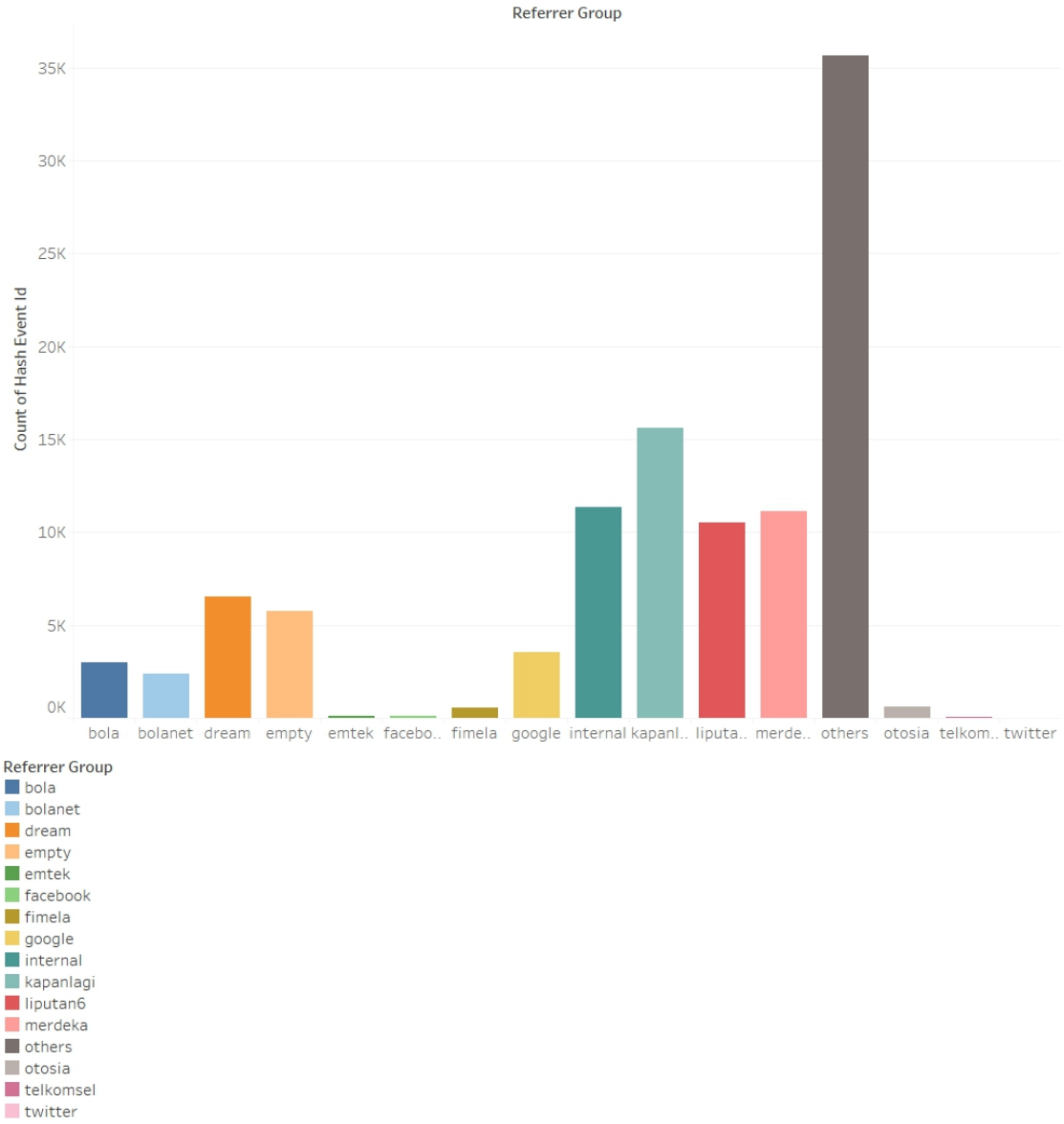


It's almost split evenly, but it did prove that embedded video is a thing. **People watch embedded video when they access some website or their social media.**

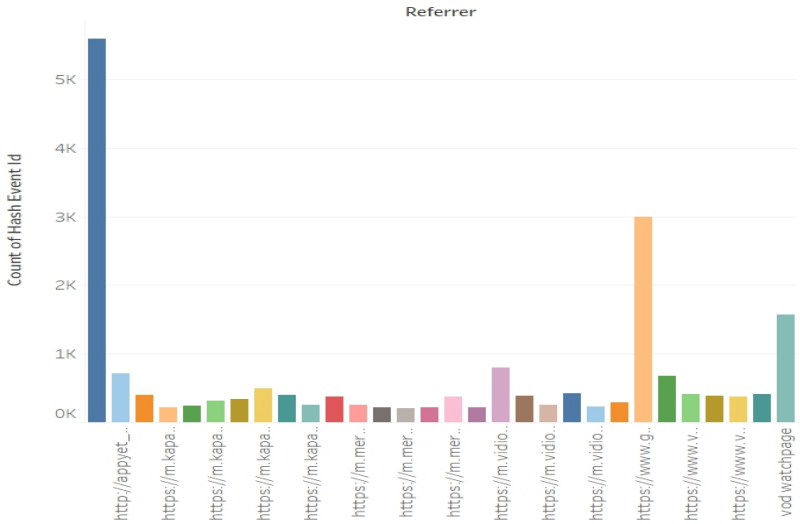
Embedded video is like when you open and explore your Facebook, then saw some video of news or ads or a piece of soap opera. Seems familiar right? Mostly because if we access social media it is unavoidable. Right now we live in the age of technology and meme culture. Everything can be spread quickly. Watching memes in video format is a norm compared the time in the past.

Ads in video format probably the main thing now and compared to static ads, watching some videos indeed is more entertaining. I suspect some of the embedded video is also in video on demand. **So embedded video maybe actually got selected in either internet or Vidio algorithm before.** There's a chart for content that we can check later, but now let's focus on the embedded video. And maybe can determine wether the algorithm have a hand.

We can check the embedded video with both referrer and referrer group chart. So, shall we?



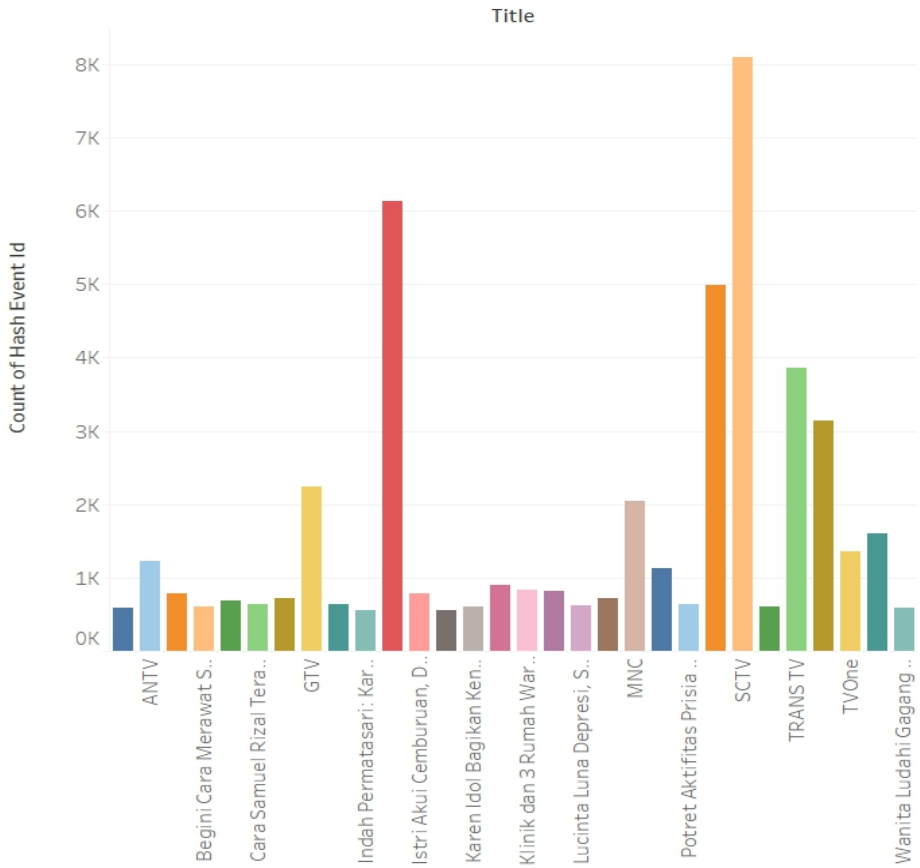
In this chart we can see several things. Many of the high numbers consist of news website, while google despite previously high number doesn't translate to this chart. So the majority of the viewers actually got referred from these news site. How about the highest number, others? I suspect this is actually a collection of several different websites that referred to the video. These smaller website is accumulated to others, thus, it is the highest referrer group. They may be a web page of news, gossip, or something. To further our understanding, let's check another chart.



Referrer

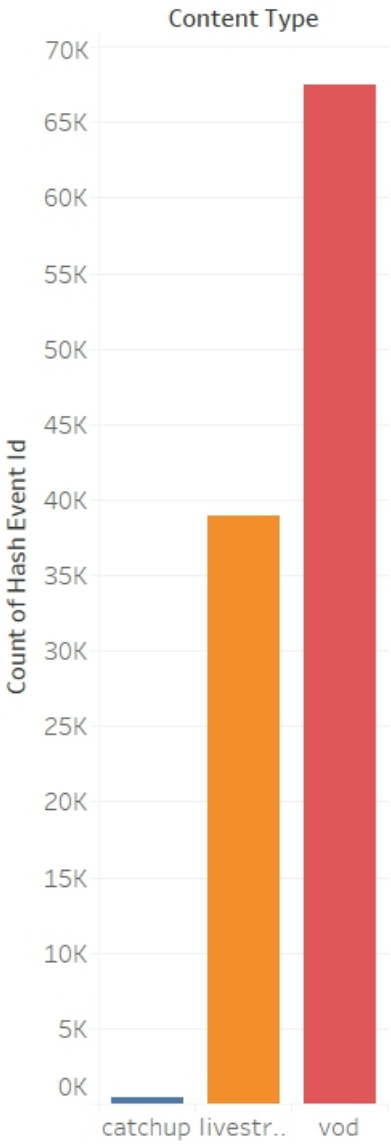
- home
- http://appyet_base/
- https://m.kapanlagi.com/
- https://m.kapanlagi.com/dangdut/berita-foto/15-foto-lawas-lucinta-luna-dari-koleksi-mantan-bos-manager-tak-tahu-karena-tak-kenal-lucinta-zaman-dulu.html?utm_source=fb&utm_medium=Post&utm_campaign=fb__adie
- https://m.kapanlagi.com/dangdut/berita-foto/78707lucinta_luna-20200215-005-kapanlagi.html
- https://m.kapanlagi.com/dangdut/berita-foto/78707lucinta_luna-20200215-032-kapanlagi.html
- https://m.kapanlagi.com/dangdut/gebby-vesta-ungkap-bayaran-yang-diterima-lucinta-luna-saat-jual-diri-di-malaysia-de7688.html?utm_source=fb&utm_medium=Post&utm_campaign=fb__adie
- https://m.kapanlagi.com/dangdut/matcont-sering-disebut-aslinya-perempuan-abash-pacar-lucinta-luna-nekat-renang-bertelanjang-dada-29f013.html?utm_source=fb&utm_medium=post&utm_campaign=fb__ajeng
- https://m.kapanlagi.com/foto/berita-foto/indonesia/8-potret-lucinta-luna-saat-ditangkap-aparat-terkait-narkoba-wajah-sedih-tanpa-wig-dan-make-up.html?utm_source=fb&utm_medium=Post&utm_campaign=fb__ajeng
- https://m.kapanlagi.com/foto/berita-foto/indonesia/78611lucinta_luna-20200211-019-non_fotografer_kly.html
- https://m.merdeka.com/
- https://m.merdeka.com/artis/8-potret-rumah-manohara-mewah-dan-ruang-tamu-nya-seluas-lapangan-bola.html?utm_source=Facebook&utm_medium=post&utm_campaign=fb__yudi
- https://m.merdeka.com/artis/dulu-jadi-babysitter-di-kampung-ini-potret-rumah-baru-evi-masamba-setelah-sukses.html?utm_source=FB&utm_medium=post&utm_campaign=FB_rdks
- https://m.merdeka.com/artis/potret-dulu-dan-sekarang-rumah-lesti-dulu-temboknya-anyaman-bambu-kini-berlantai-dua.html
- https://m.merdeka.com/artis/wendy-cagur-dan-kawan-kawan-lagi-ngaleng-digerebek-polisi.html?utm_source=FB&utm_medium=post&utm_campaign=FB_rdks
- https://m.merdeka.com/trending/bunda-sarita-pernah-nikah-sama-pria-blasteran-jerman-ini-6-potret-anaknya-kln.html?utm_source=Facebook&utm_medium=post&utm_campaign=fb__yudi
- https://m.merdeka.com/trending/lama-menghilang-begini-kondisi-chef-haryo-sekarang-yang-mantap-berhijrah.html
- https://m.vidio.com/live
- https://m.vidio.com/live/204-sctv
- https://m.vidio.com/live/205-indosiar
- https://m.vidio.com/live/665-rcti
- https://m.vidio.com/live/733-trans-tv
- https://www.google.co.id/
- https://www.google.com/
- https://www.vidio.com/live
- https://www.vidio.com/live/665-rcti
- https://www.vidio.com/live/733-trans-tv
- https://www.vidio.com/live/734-trans7
- https://www.youtube.com/
- vod watchpage

Let's just focus on how many webpage in the chart rather than the highest number. For comparison, the chart below is from the title.



Title
- 4 Januari, Hari Bahagia dan Paling Sedih Rizky Febian
- ANTV
- Babak Baru Daus Mini Vs Daus Mini 'KW'
- Begini Cara Merawat Sepatu Sneakers Supaya Selalu Terlihat Baru
- Begini Reaksi Ular Kobra Saat Diajak Breakdance
- Cara Samuel Rizal Terapkan Hidup Sehat Pada Anak
- Dony Pedro \King of The King\ Kerajaan Baru di Tangerang
- GTV
- Ibunda Annisa Bahar Meninggal Dunia
- Indah Permatasari: Karena Mimpi Aku Hidup
- Indosiar
- Istri Akui Cemburuan, Daus Mini: Coverboy Sih Nggak ya
- Jokowi Minta Kalkulasi Dampak Virus Corona Terhadap Perekonomian Indonesia
- Karen Idol Bagikan Kenangan dengan Sang Anak di Instagram
- Kesal Obat Habis, Pria Wuhan Sengaja Batuk ke Arah Perawat
- Klinik dan 3 Rumah Warga Pancoran Ludes Terbakar
- Kompas TV
- Lucinta Luna Depresi, Sempat Ingin Bunuh Diri 3 Kali
- Metro TV
- MNC
- NET TV
- Potret Aktifitas Prisia Nasution Saat Jelajahi Alam
- RCTI
- SCTV
- Seorang Ayah Selamatkan Putrinya dari Lintasan Kereta Api
- TRANS TV
- Trans7
- TVOne
- Video Polisi Tes Kandungan Sabu Cair dalam Mainan Anak-anak
- Wanita Ludahi Gagang Pintu Apartemennya, Diduga Sebar Virus Corona

Based on the charts above, we can conclude that theres actually many programs or videos that is watched by the viewers, and they can be different as night and day. Even if the charts only limited to several top numbers, it shows that, yes, there's maybe some algorithm at play here. The referrer is dominated by several website at the top, but it's probable they have many visitors and clicks. Several website also have many pages that popular and specifically accessed, leading to the viewers watching the videos. For the last argument, we check the content type chart.



With this, it's clear that the content only have 3 types. With those many livestream and vod, it means that the video watched are connected with the referrals. The video maybe a short clip, ads, music, news, or anything else, but it is certainly connected with how the viewers used the phone.

The viewers mostly access the content with mobile phone, they often access the video from the website, there are diverse type of viewers, there are many type of programs that is watched, and most of the time viewers are watching free contents so they have ads.